



8.3.2 Existence of A-conjugate search directions



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The big question left dangling at the end of the last unit was whether there exists a direction $p^{(k)}$ that is A-orthogonal to all previous search directions and that is not orthogonal to $r^{(k)}$. Let us examine this:

- Assume that all prior search directions $p^{(0)}, \dots, p^{(k-1)}$ were A-conjugate.
- Consider all vectors $p \in \mathbb{R}^n$ that are A-conjugate to $p^{(0)}, \dots, p^{(k-1)}$. A vector p has this property if and only if $p \perp \text{Span}(Ap^{(0)}, \dots, Ap^{(k-1)})$.
- For $p \perp \text{Span}(Ap^{(0)}, \dots, Ap^{(k-1)})$ we notice that

$$p^T r^{(k)} = p^T (b - Ax^{(k)}) = p^T (b - AP^{(k-1)} a^{(k-1)})$$

where we recall from (8.3.2) that

$$P^{(k-1)} = \begin{pmatrix} p^{(0)} & \dots & p^{(k-1)} \end{pmatrix} \text{ and } a^{(k-1)} = \begin{pmatrix} \alpha_0 \\ \vdots \\ \alpha_{k-1} \end{pmatrix}.$$

- If all vectors p that are A-conjugate to $p^{(0)}, \dots, p^{(k-1)}$ are orthogonal to the current residual, $p^T r^{(k)} = 0$ for all p with $P^{(k-1)T} Ap = 0$, then

$$0 = p^T b - pAP^{(k-1)} a^{(k-1)} = p^T b \text{ for all } p \perp \text{Span}(Ap^{(0)}, \dots, Ap^{(p-1)}).$$

Let's think about this: b is orthogonal to all vectors that are orthogonal to $\text{Span}(Ap^{(0)}, \dots, Ap^{(p-1)})$. This means that

$$b \in \text{Span}(Ap^{(0)}, \dots, Ap^{(k-1)}).$$

- Hence $b = AP^{(k-1)} z$ for some $z \in \mathbb{R}^k$. It also means that $x = P^{(k-1)} z$ solves $Ax = b$.
- We conclude that our method must already have found the solution since $x^{(k)}$ minimizes $f(x)$ over all vectors in $\text{Span}(p^{(0)}, \dots, p^{(k-1)})$. Thus $Ax^{(k)} = b$ and $r^{(k)} = 0$.

We conclude that there exist descent methods that leverage A-conjugate search directions as described in Figure 8.3.1.3. The question now is how to find a new A-conjugate search direction at every step.